

ECON 609: Perturbation

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Lecture # 5

Introduction

- ▶ Remember that we want to solve a functional equation of the form:

$$\mathcal{H}(d) = \mathbf{0}$$

for an unknown decision rule d .

- ▶ Perturbation solves the problem by specifying:

$$d^n(x, \theta) = \sum_{i=0}^n \theta_i (x - x_0)^i$$

- ▶ We want find coefficients θ_i 's such that $\mathcal{H}(d^n) \approx \mathbf{0}$.
- ▶ Inherently local approximation. However, often good global properties.

Motivation

- ▶ Many complicated mathematical problems have:
 - (1) either a particular case
 - (2) or a related problem that is easy to solve.
- ▶ Often, we can use the solution of the simpler problem as a building block of the general solution.
- ▶ Very successful in physics.

Simplest Perturbation

- ▶ What is $\sqrt{26}$?
- ▶ Without your calculator, it is a boring arithmetic calculation.
- ▶ But note that:

$$\sqrt{26} = \sqrt{25(1 + 0.04)} = 5 * \sqrt{1.04} \approx 5 * 1.02 = 5.1$$

- ▶ Exact solution is 5.099.
- ▶ We have solved a much simpler problem ($\sqrt{25}$) and added a small coefficient to it.
- ▶ More in general: $\sqrt{y} = \sqrt{x(1 + \epsilon)} = x\sqrt{1 + \epsilon}$ where ϵ is the perturbation parameter.

Applications to Economics

- ▶ Judd and Guu (1993) showed how to apply it to economic problems.
- ▶ Recently, perturbation methods have been gaining much popularity.
- ▶ In particular, second- and third-order approximations are easy to compute.
- ▶ A first-order perturbation theory and linearization deliver the same output.

A Simple Example: A Basic RBC

► Model:

$$\max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \log c_t$$

$$s.t. \ c_t + k_{t+1} = e^{z_t} k_t^\alpha + (1 - \delta)k_t, \quad \forall t > 0$$

$$z_t = \rho z_{t-1} + \sigma \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, 1)$$

► Equilibrium conditions:

$$\frac{1}{c_t} = \beta \mathbb{E}_t \frac{1}{c_{t+1}} (1 + \alpha e^{z_{t+1}} k_{t+1}^{\alpha-1} - \delta)$$

$$c_t + k_{t+1} = e^{z_t} k_t^\alpha + (1 - \delta)k_t$$

$$z_t = \rho z_{t-1} + \sigma \epsilon_t$$

Computing a Solution

- ▶ The previous problem does not have a known "paper and pencil" solution except when $\delta = 1$.
- ▶ Equilibrium condition with $\delta = 1$:

$$\frac{1}{c_t} = \beta \mathbb{E}_t \frac{\alpha e^{z_{t+1}} k_{t+1}^{\alpha-1}}{c_{t+1}}$$

$$c_t + k_{t+1} = e^{z_t} k_t^\alpha$$

$$z_t = \rho z_{t-1} + \sigma \epsilon_t$$

- ▶ By "guess and verify":

$$c_t = (1 - \alpha\beta) e^{z_t} k_t^\alpha$$

$$k_{t+1} = \alpha\beta e^{z_t} k_t^\alpha$$

Another Way to Solve the Problem

- ▶ Now let us suppose that we need to compute the RBC not knowing "guess and verify."
- ▶ What are we searching for? A decision rule for consumption:

$$c_t = c(k_t, z_t)$$

and another one for capital:

$$k_{t+1} = k(k_t, z_t)$$

Equilibrium Conditions

- ▶ We substitute in the equilibrium conditions the law of motion for technology.
- ▶ And we write the decisions rules explicitly as function of the states.
- ▶ We obtain the system of functional equations

Main Idea of Perturbation

- ▶ Transform the problem rewriting it in terms of a small perturbation parameter.
- ▶ Solve the new problem for a particular choice of the perturbation parameter.
- ▶ There are different ways to do so.
- ▶ Use the previous solution to approximate the solution of the original problem.

A Perturbation Approach

- ▶ Hence, we want to transform the problem.
- ▶ Which perturbation parameter? Standard deviation σ .
- ▶ Set $\sigma = 0 \rightarrow$ deterministic model, $z_t = 0$ and $e^{z_t} = 1$.
- ▶ We know how to solve the deterministic steady state.

A Parameterized Decision Rule

- ▶ We search for decision rules:

$$c_t = c(k_t, z_t; \sigma)$$

$$k_{t+1} = k(k_t, z_t; \sigma)$$

- ▶ Note new parameter σ .
- ▶ We are building a local approximation around $\sigma = 0$.

Taylor's Theorem

- Equilibrium conditions:

$$\mathbb{E}_t \left(\frac{1}{c(k_t, z_t; \sigma)} - \beta \frac{\alpha e^{\rho z_t + \sigma \epsilon_{t+1}} k(k_t, z_t; \sigma)^{\alpha-1}}{c(k(k_t, z_t; \sigma), \rho z_t + \sigma \epsilon_{t+1}; \sigma)} \right) = 0$$
$$c(k_t, z_t; \sigma) + k(k_t, z_t; \sigma) - e^{z_t} k_t^\alpha = 0$$

- We will take derivatives with respect to k_t , z_t , and σ .
- Apply Taylor's theorem to build solution around deterministic steady state. How?

Asymptotic Expansion I

$$\begin{aligned}c_t = & c(k_t, z_t; \sigma)|_{k,0,0} = c(k, 0; 0) \\& + c_k(k, 0; 0)(k_t - k) + c_z(k, 0; 0)z_t + c_\sigma(k, 0; 0)\sigma \\& + \frac{1}{2}c_{kk}(k, 0; 0)(k_t - k)^2 + \frac{1}{2}c_{kz}(k, 0; 0)(k_t - k)z_t \\& + \frac{1}{2}c_{k\sigma}(k, 0; 0)(k_t - k)\sigma + \frac{1}{2}c_{zk}(k, 0; 0)z_t(k_t - k) \\& + \frac{1}{2}c_{zz}(k, 0; 0)z_t^2 + \frac{1}{2}c_{z\sigma}(k, 0; 0)z_t\sigma \\& + \frac{1}{2}c_{\sigma k}(k, 0; 0)\sigma(k_t - k) + \frac{1}{2}c_{\sigma z}(k, 0; 0)\sigma z_t \\& + \frac{1}{2}c_{\sigma\sigma}(k, 0; 0)\sigma^2 + \dots\end{aligned}$$

Asymptotic Expansion II

$$\begin{aligned}k_{t+1} = & k(k_t, z_t; \sigma)|_{k,0,0} = k(k, 0; 0) \\& + k_k(k, 0; 0)(k_t - k) + k_z(k, 0; 0)z_t + k_\sigma(k, 0; 0)\sigma \\& + \frac{1}{2}k_{kk}(k, 0; 0)(k_t - k)^2 + \frac{1}{2}k_{kz}(k, 0; 0)(k_t - k)z_t \\& + \frac{1}{2}k_{k\sigma}(k, 0; 0)(k_t - k)\sigma + \frac{1}{2}k_{zk}(k, 0; 0)z_t(k_t - k) \\& + \frac{1}{2}k_{zz}(k, 0; 0)z_t^2 + \frac{1}{2}k_{z\sigma}(k, 0; 0)z_t\sigma \\& + \frac{1}{2}k_{\sigma k}(k, 0; 0)\sigma(k_t - k) + \frac{1}{2}k_{\sigma z}(k, 0; 0)\sigma z_t \\& + \frac{1}{2}k_{\sigma\sigma}(k, 0; 0)\sigma^2 + \dots\end{aligned}$$

Comment on Notation

- From now on, to save on notation,

$$F(k_t, z_t; \sigma) = \mathbb{E}_t \left[\frac{1}{c(k_t, z_t; \sigma)} - \beta \frac{\alpha e^{\rho z_t + \sigma \epsilon_{t+1}} k(k_t, z_t; \sigma)^{\alpha-1}}{c(k(k_t, z_t; \sigma), \rho z_t + \sigma \epsilon_{t+1}; \sigma)} \right] = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

- Using functional equations' notation:

$$\begin{aligned} F(k_t, z_t; \sigma) &= \mathcal{H}(c_t, c_{t+1}, k_t, k_{t+1}, z_t; \sigma) \\ &= \mathcal{H}(c(k_t, z_t; \sigma), c(k(k_t, z_t; \sigma), z_{t+1}; \sigma), k_t, k(k_t, z_t; \sigma), z_t; \sigma) \end{aligned}$$

- We will use $\mathcal{H}_i$ to represent the partial derivative of $\mathcal{H}$ with respect to the i th component.

Zeroth-Order Approximation

- First, we evaluate $\sigma = 0$:

$$F(k_t, 0; 0) = 0$$

- Steady state:

$$\frac{1}{c} = \beta \frac{\alpha k^{\alpha-1}}{c}$$

or

$$1 = \alpha \beta k^{\alpha-1}$$

- Then:

$$c = c(k, 0; 0) = (\alpha \beta)^{\frac{\alpha}{1-\alpha}} - (\alpha \beta)^{\frac{1}{1-\alpha}}$$

$$k = k(k, 0; 0) = (\alpha \beta)^{\frac{1}{1-\alpha}}$$

First-Order Approximation

- ▶ We take derivatives of $F(k_t, z_t; \sigma)$ around $(k, 0; 0)$.

- ▶ With respect to k_t :

$$F_k(k, 0; 0) = 0$$

- ▶ With respect to z_t :

$$F_z(k, 0; 0) = 0$$

- ▶ With respect to σ :

$$F_\sigma(k, 0; 0) = 0$$

Solving the System I

- ▶ Remember that:

$$F(k_t, z_t; \sigma) = \mathcal{H}(c(k_t, z_t; \sigma), c(k(k_t, z_t; \sigma), z_{t+1}; \sigma), k_t, k(k_t, z_t; \sigma), z_t; \sigma) = 0$$

- ▶ Because $F(k_t, z_t; \sigma)$ must be equal to zero for any possible values of k_t, z_t , and σ , the derivatives of any order of F must also be zero.

- ▶ Then:

$$F_k(k, 0; 0) = \mathcal{H}_1 c_k + \mathcal{H}_2 c_k k_k + \mathcal{H}_3 + \mathcal{H}_4 k_k = 0$$

$$F_z(k, 0; 0) = \mathcal{H}_1 c_z + \mathcal{H}_2 (c_k k_z + c_k \rho) + \mathcal{H}_4 k_z + \mathcal{H}_5 = 0$$

$$F_\sigma(k, 0; 0) = \mathcal{H}_1 c_\sigma + \mathcal{H}_2 (c_k k_\sigma + c_\sigma) + \mathcal{H}_4 k_\sigma + \mathcal{H}_6 = 0$$

Solving the System II

- ▶ A quadratic system:

$$F_k(k, 0; 0) = \mathcal{H}_1 c_k + \mathcal{H}_2 c_k k_k + \mathcal{H}_3 + \mathcal{H}_4 k_k = 0$$

$$F_z(k, 0; 0) = \mathcal{H}_1 c_z + \mathcal{H}_2 (c_k k_z + c_k \rho) + \mathcal{H}_4 k_z + \mathcal{H}_5 = 0$$

from which we solve for c_k , c_z , k_k , and k_z .

- ▶ Procedures to solve quadratic systems: Blanchard and Kahn (1980), Uhlig (1999), Sims (2000), Klein (2000)
- ▶ All of them are equivalent.
- ▶ Recall: linearization lecture note.

Solving the System III

- ▶ Also, note that:

$$F_{\sigma}(k, 0; 0) = \mathcal{H}_1 c_{\sigma} + \mathcal{H}_2(c_k k_{\sigma} + c_{\sigma}) + \mathcal{H}_4 k_{\sigma} + \mathcal{H}_6 = 0$$

is a linear, and homogeneous system in c_{σ} and k_{σ} .

- ▶ Hence, $c_{\sigma} = k_{\sigma} = 0$.
- ▶ This means that the system is certainty equivalent.
- ▶ Interpretation: no precautionary behavior.

Second-Order Approximation

- ▶ We take second-order derivatives of $F(k_t, z_t; \sigma)$ around $(k, 0; 0)$:

$$F_{kk}(k, 0; 0) = 0, \quad F_{kz}(k, 0; 0) = 0, \quad F_{k\sigma}(k, 0; 0) = 0$$

$$F_{zz}(k, 0; 0) = 0, \quad F_{z\sigma}(k, 0; 0) = 0, \quad F_{\sigma\sigma}(k, 0; 0) = 0$$

- ▶ Remember Young's theorem.
- ▶ We substitute the coefficients that we already know.
- ▶ We get linear system of equations after substitution.
- ▶ What will be the cross derivative terms regarding $k\sigma$ and $z\sigma$?
e.g. $c_{k\sigma}, k_{z\sigma}$

Correction for Risk

- ▶ We have a term in σ^2 .
- ▶ Captures precautionary behavior.
- ▶ We do not have certainty equivalence anymore.
- ▶ Important advantage of second-order approximation.
- ▶ We can continue the iteration for as long as we want for higher-order terms.
- ▶ Often, a few iterations will be enough.

In Practice

- ▶ We do all these approximations with a computer.
- ▶ For the first-, second-, and third-order approximation, we can use Matlab and Dynare.
- ▶ Perturbation is a local method, but we can check numerically whether perturbations have good non-local behavior.

Perturbing the Value Function

- ▶ We worked with the equilibrium conditions of the model.
- ▶ Sometimes we may want to perform a perturbation on the value function formulation of the problem.
- ▶ Possible reasons:
 - ▶ Gain insight
 - ▶ Difficulty in using equilibrium conditions
 - ▶ Evaluate welfare

Basic Problem

- Imagine that we have:

$$V(k_t, z_t) = \max_{c_t} \left[(1 - \beta) \frac{c_t^{1-\gamma}}{1-\gamma} + \beta \mathbb{E}_t V(k_{t+1}, z_{t+1}) \right]$$

$$s.t. \ c_t + k_{t+1} = e^{z_t} k_t^\theta + (1 - \delta) k_t$$

$$z_t = \lambda z_{t-1} + \sigma \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, 1)$$

- Write it as:

$$V(k_t, z_t; \chi) = \max_{c_t} \left[(1 - \beta) \frac{c_t^{1-\gamma}}{1-\gamma} + \beta \mathbb{E}_t V(k_{t+1}, z_{t+1}; \chi) \right]$$

$$s.t. \ c_t + k_{t+1} = e^{z_t} k_t^\theta + (1 - \delta) k_t$$

$$z_t = \lambda z_{t-1} + \chi \sigma \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, 1)$$

Alternative

- ▶ Another way to write the value function is:

$$V(k_t, z_t; \chi) = \max_{c_t} \left[(1 - \beta) \frac{c_t^{1-\gamma}}{1 - \gamma} + \beta \mathbb{E}_t V(e^{z_t} k_t^\theta + (1 - \delta)k_t - c_t, \lambda z_t + \chi \sigma \epsilon_{t+1}; \chi) \right]$$

- ▶ This form makes the dependence in the next period's states explicit.
- ▶ The solution of this problem is value function $V(k_t, z_t; \chi)$ and a policy function $c(k_t, z_t; \chi)$.

Expanding the Value Function

- The second-order Taylor approximation of the value function around the deterministic steady state $(k_{ss}, 0; 0)$ is:

$$\begin{aligned} V(k_t, z_t; \chi) \approx & V_{ss} + V_{1,ss}(k_t - k_{ss}) + V_{2,ss}z_t + V_{3,ss}\chi \\ & + \frac{1}{2} V_{11,ss}(k_t - k_{ss})^2 + \frac{1}{2} V_{12,ss}(k_t - k_{ss})z_t + \frac{1}{2} V_{13,ss}(k_t - k_{ss})\chi \\ & + \frac{1}{2} V_{21,ss}z_t(k_t - k_{ss}) + \frac{1}{2} V_{22,ss}z_t^2 + \frac{1}{2} V_{23,ss}z_t\chi \\ & + \frac{1}{2} V_{31,ss}\chi(k_t - k_{ss}) + \frac{1}{2} V_{32,ss}\chi z_t + \frac{1}{2} V_{33,ss}\chi^2 \end{aligned}$$

where

$$V_{ss} = V(k_{ss}, 0; 0)$$

$$V_{i,ss} = V_i(k_{ss}, 0; 0) \quad \text{for } i = \{1, 2, 3\}$$

$$V_{ij,ss} = V_{ij}(k_{ss}, 0; 0) \quad \text{for } i, j = \{1, 2, 3\}$$

Expanding the Value Function

- By certainty equivalence, we can show that

$$V_{3,ss} = V_{13,ss} = V_{23,ss} = 0$$

- Taking advantage of the equality of cross-derivatives, and setting $\chi = 1$:

$$\begin{aligned} V(k_t, z_t; 1) &\approx V_{ss} + V_{1,ss}(k_t - k_{ss}) + V_{2,ss}z_t \\ &\quad + \frac{1}{2}V_{11,ss}(k_t - k_{ss})^2 + \frac{1}{2}V_{22,ss}z_t^2 \\ &\quad + V_{12,ss}(k_t - k_{ss})z_t + \frac{1}{2}V_{33,ss} \end{aligned}$$

- Note that $V_{33,ss} \neq 0$.

The Welfare Cost of the Business Cycle

- ▶ An advantage of performing the perturbation on the value function is that we have evaluation of welfare readily available.

- ▶ Note that at $z_{ss} = 0$, we have:

$$V(k_{ss}, 0; \chi = 1) \approx V_{ss} + \frac{1}{2} V_{33,ss}$$

- ▶ $\frac{1}{2} V_{33,ss}$ is a measure of the welfare cost of the business cycle.
- ▶ Linearization cannot capture this term: correction due to aggregate fluctuation.

In Consumption Terms

- ▶ We know that $V_{ss} = \frac{c_{ss}^{1-\gamma}}{1-\gamma}$.
- ▶ We can compute the decrease in consumption τ that will make the household indifferent between consuming $(1 - \tau)c_{ss}$ units per period with certainty vs. c_t units with uncertainty.

$$\frac{(c_{ss}(1 - \tau))^{1-\gamma}}{1 - \gamma} = \frac{c_{ss}^{1-\gamma}}{1 - \gamma} + \frac{1}{2} V_{33,ss}$$
$$\therefore \tau = 1 - \left[1 + \frac{1 - \gamma}{c_{ss}^{1-\gamma}} \frac{1}{2} V_{33,ss} \right]$$